FR800x Datasheet

Bluetooth Low Energy SOC with SIG Mesh integrated

Version:v0.3.19

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FREQCHIP

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Description

FR800x is a family of SOC (system on chip) for rapid development of Bluetooth Low Energy related products. It contains Bluetooth V5.3 (LE Mode) fully compliant system with Freqchip designed firmware and software stack. Users can develop various applications based on embedded 32-bits high performance MCU.

The Bluetooth Smart firmware includes the L2CAP service layer protocols, Security Manager (SM), Attribute Protocol (ATT), the Generic Attribute Profile (GATT) and the Generic Access Profile (GAP). Furthermore, application profiles such as Proximity, Health Thermometer, Heart Rate, Blood Pressure, Glucose, Human Interface Device (HID) and SDK (include drivers, OS API, etc.) are supported. The SDK has integrated SIG Mesh for networking application.

With Freqchip's innovational technology, FR800x integrates, PMU, QSPI flash ROM with XIP mode, I2C, UART, GPIO, ADC, PWM all in a single chip, which provides customer with:

1. competitive power consumption
2. stable connection
3. low-cost BOM

Features

CPU and On-chip Memory

- Embedded 32-bits CPU
    - Running at a frequency of up to 96MHz
- Memory
    - 512KB/1MB/4MB/8MB/16MB Flash for user space software and data
    - up to 56KB SRAM
    - Internal mask 128KB ROM
        - BOOT Code
        - Controller protocol stack
    - ROM Software:
        - BLE Profile & Protocol: GATT, LM, LC
        - Driver API
        - SIG Mesh

Bluetooth

- Compliant with Bluetooth Specification V5.3 LE
- support 2M, 1M, 500K and 125K data rate

PMU

- Integrated DC-DC Regulator

Interface

- Digital Interface:
    - GPIO
    - Timer *2
    - Efuse 128bit
    - SPIM *2
    - SPIS
    - UART(FIFO Depth 16/32)
    - I2C(FIFO Depth 8/32)
    - PWM *8
    - PDM
    - USB OTG
    - 8080 parallel interface
- Analog Interface:
    - 8-channel 10bit SAR ADC
    - Maximum sampling data rate 1Mbp
    - Sampling trigger follows PWM pulse

Operating Conditions:

- Operating ambient temperature: -40°C ~ +85°C

Applications

- Smart Mouse
- Bluetooth Wearable
- SIG Mesh Application
- Smart Locks
- Domestic Applianc

Ordering Information

Part Number	Ambient temperature	Embedded Flash	Embedded Psram	Package	Size	MSL
FR8003A	-40°C ~ +85°C	512KB	0	QFN20	3.0×3.0×0.75, 0.4pitch	MSL3
FR8003D	-40°C ~ +85°C	1MB	0	QFN20	3.0×3.0×0.75, 0.4pitch	MSL3
FR8008A	-40°C ~ +85°C	512KB	0	QFN48	6.0×6.0×0.75, 0.4pitch	MSL3
FR8008G	-40°C ~ +85°C	8MB	0	QFN40	5.0×5.0×0.75, 0.4pitch	MSL3
FR8008AP	-40°C ~ +85°C	512KB	2MB	QFN40	5.0×5.0×0.75, 0.4pitch	MSL3
FR8008FP	-40°C ~ +85°C	4MB	2MB	QFN40	5.0×5.0×0.90, 0.4pitch	MSL3
FR8008GP-P	-40°C ~ +85°C	8MB	2MB	QFN40	5.0×5.0×0.90, 0.4pitch	MSL3
FR8008HP	-40°C ~ +85°C	16MB	2MB	QFN40	5.0×5.0×0.90, 0.4pitch	MSL3

1. Hardware Details

1.1 Block Diagram

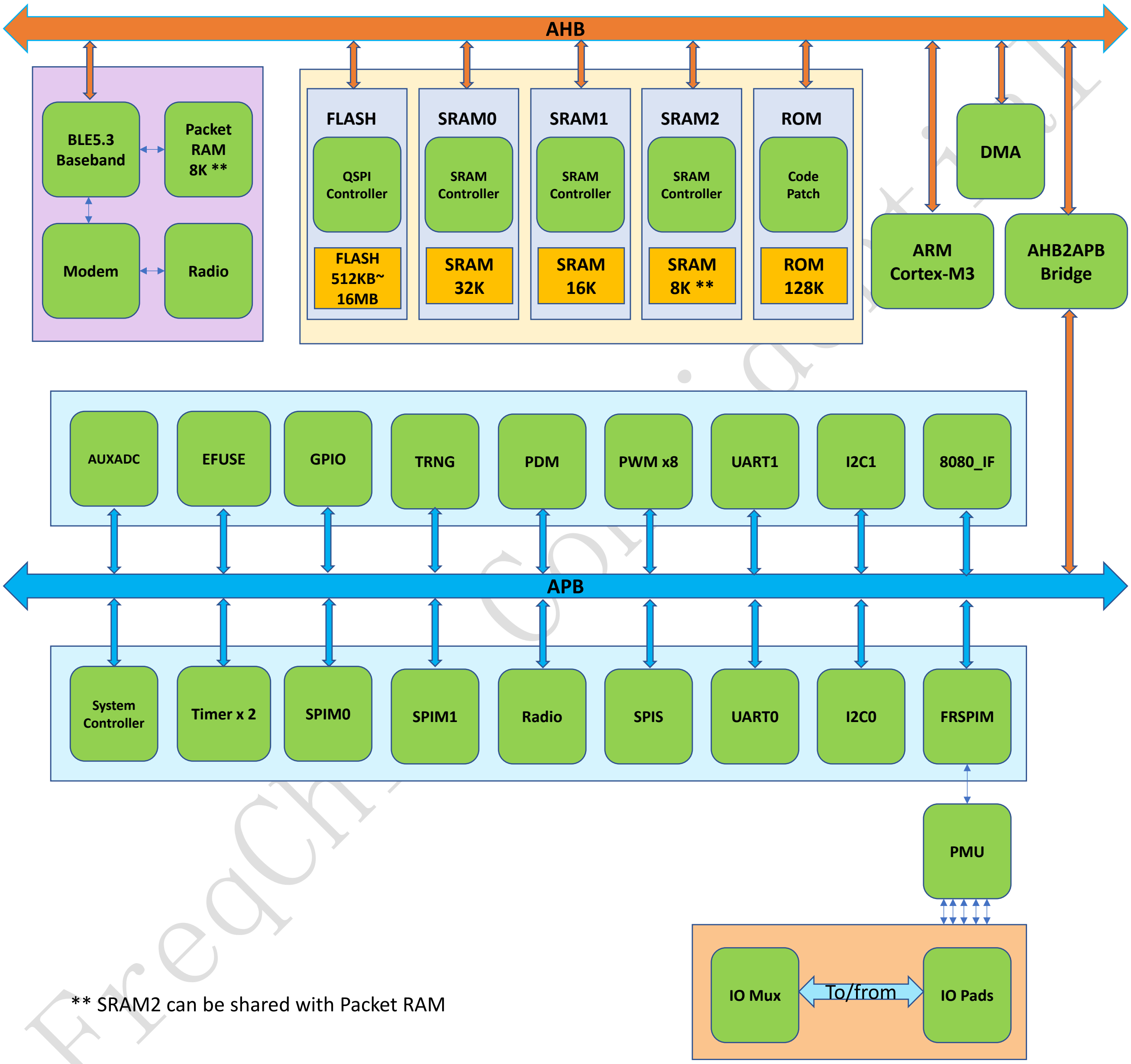


Figure 1-1 Block Diagram

1.2 Hardware Resources

Hardware resources of FR800x series are shown in the following table:

Table 1-1 FR800x Hardware Resources

Part Number	Frequency	GPIO	HD_ONKEY	FLASH	SRAM	PSRAM	Timer	Watchdog	RTC	UART	I2C	SPIM	SPIS	I2S	EFUSE	8080	PDM	TRANG	PWM	USB+ USB(GPIO)	LCD	SARADC
FR8003A	96M	12	Y	512KB	56K	0	2	1	1	2	2	2	1	1	1	1	1	1	8	0+1	1	4
FR8003D	96M	12	N	1MB	56K	0	2	1	1	2	2	2	1	1	1	1	1	1	8	0+1	1	4
FR8008A	96M	34	N	512KB	56K	0	2	1	1	2	2	2	1	1	1	1	1	1	8	1+1	1	8
FR8008G	96M	28	N	8M	56K	0	2	1	1	2	2	2	1	1	1	1	1	1	8	0+1	1	8
FR8008AP	96M	28	N	512KB	56K	2M	2	1	1	2	2	2	1	1	1	1	1	1	8	1+1	1	8
FR8008FP	96M	28	N	4MB	56K	2M	2	1	1	2	2	2	1	1	1	1	1	1	8	0+1	1	8
FR8008GP-P	96M	28	N	8M	56K	2M	2	1	1	2	2	2	1	1	1	1	1	1	8	0+1	1	8
FR8008HP	96M	28	N	16M	56K	2M	2	1	1	2	2	2	1	1	1	1	1	1	8	0+1	1	8

1.3 Bluetooth Radio

- On-chip balun (50Ω impedance in TX and RX modes)
- No external trimming is required in production
- Qualified to Bluetooth v5.3 LE specification
- -20dBm ~ 10dBm RF transmit power
- -97dBm (1M) receiver sensitivity
- Integrated channel filters
- Digital demodulator for improved sensitivity and co-channel rejection
- Real time digitized RSSI

1.4 Bluetooth Controller

- All device classes support (Broadcaster, Central, Observer, Peripheral)
- All packet types (Advertising / Data / Control)
- Encryption (AES / CCM)
- Bit stream processing (CRC, Whitening)
- Frequency hopping calculation
- Low power modes supporting internal 32.0 kHz RC oscillator or external 32.768 kHz Crystal oscillator (FR8008A only)
- Supports power down of the baseband during the protocol's idle periods

1.5 Peripheral Interfaces

- UART port for Debugging and AT Commands
- I2C interface to support external EEPROM or other devices (like G-SENSOR)
- Up to 34 general purpose IOs (34 IOs can be set in interrupt mode)
- General purpose 10-bit ADC used for basic analog signal measurement
- 8-channel PWM controller, with dead time control
- General purpose programmable timer for various task
- Watchdog used for tracking unexpected exception

1.6 Integrated Power Control and Regulation

- Embedded Power-On-Reset
- On-chip high efficiency switch-mode, 2.0v to 3.6v input
- On-chip Low Dropout (LDO) Linear Regulator for internal Digital, RF and Analog circuit
- Power management features include software shutdown and hardware wake-up
- Power-on-reset cell detects low supply voltage
- Internal voltage level detector

2. Hardware Information

2.1 Package

2.1.1 FR8003A Pinout

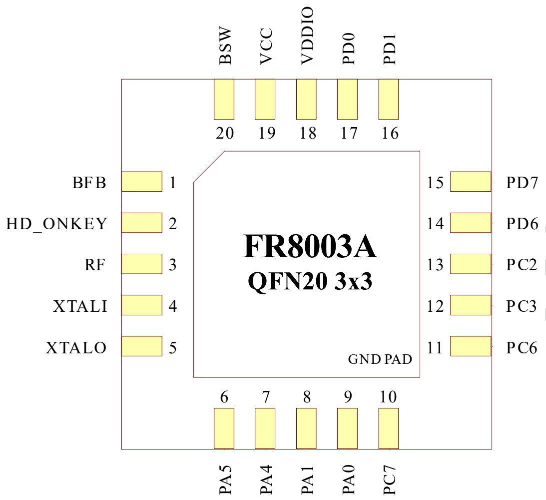


Figure 2-1 FR8003A package

2.1.2 FR8003D Pinout

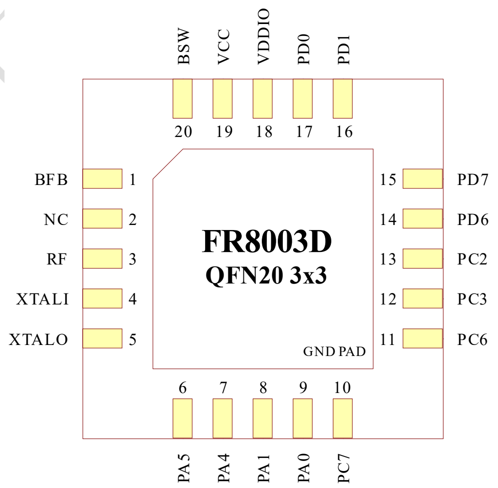


Figure 2-2 FR8003D package

2.1.3 FR8008A Pinout

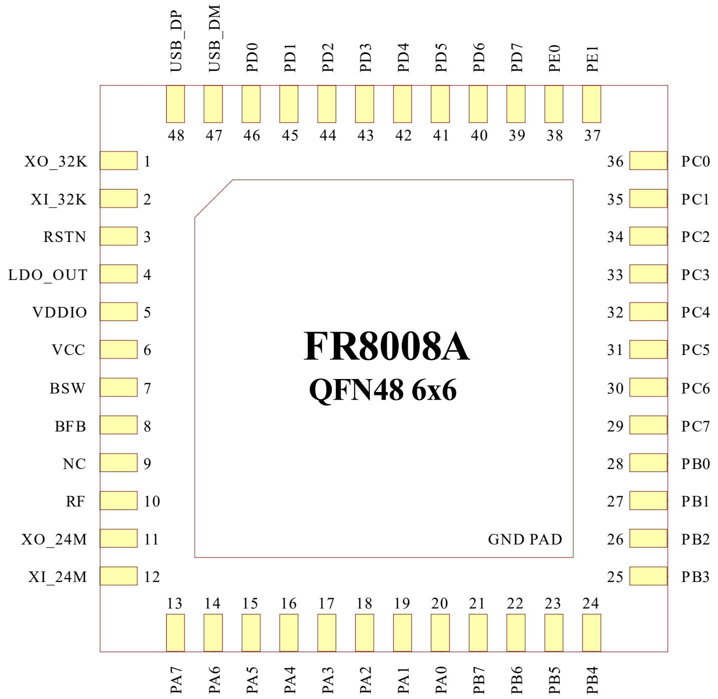


Figure 2-3 FR8008A package

2.1.4 FR8008G Pinout

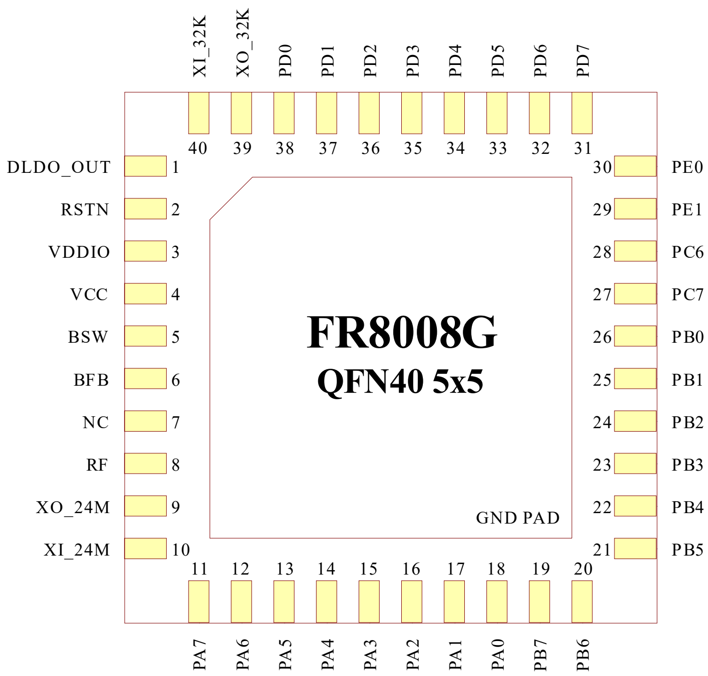


Figure 2-4 FR8008G package

2.1.5 FR8008XP(FR8008FP/FR8008GP/FR8008HP) Pinout

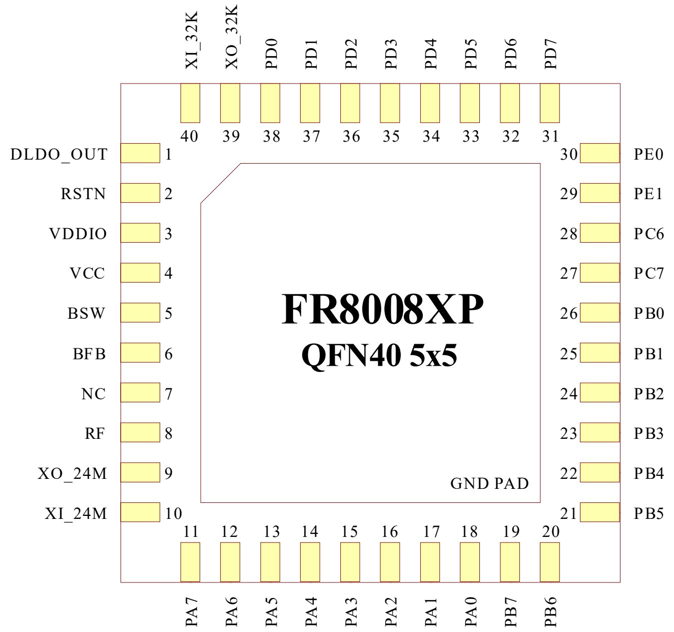


Figure 2-5 FR8008XP package

2.1.6 FR8008AP Pinout

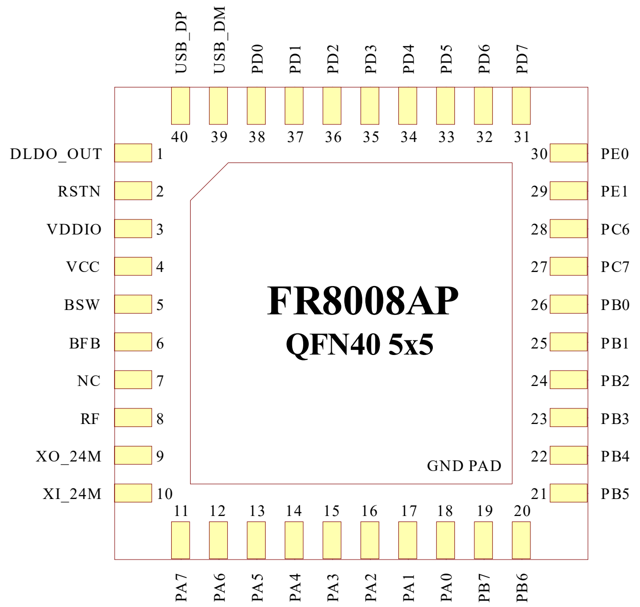
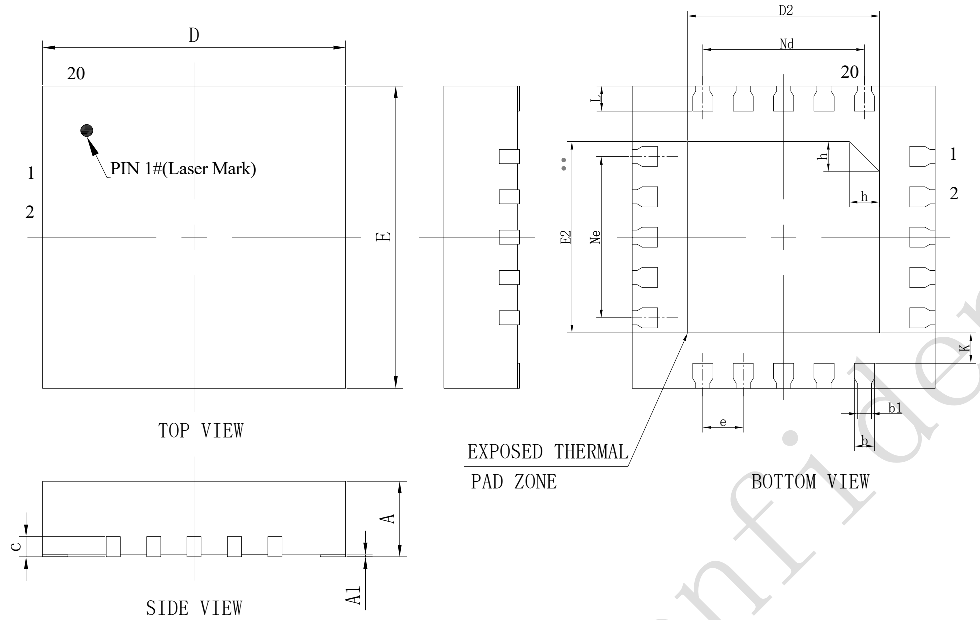


Figure 2-6 FR8008AP package

Note:
Some pins of FR8008AP are slightly different from FR8008XP, Figure 2-6shows FR8008AP pins layout

2.2 Package Physical Dimension

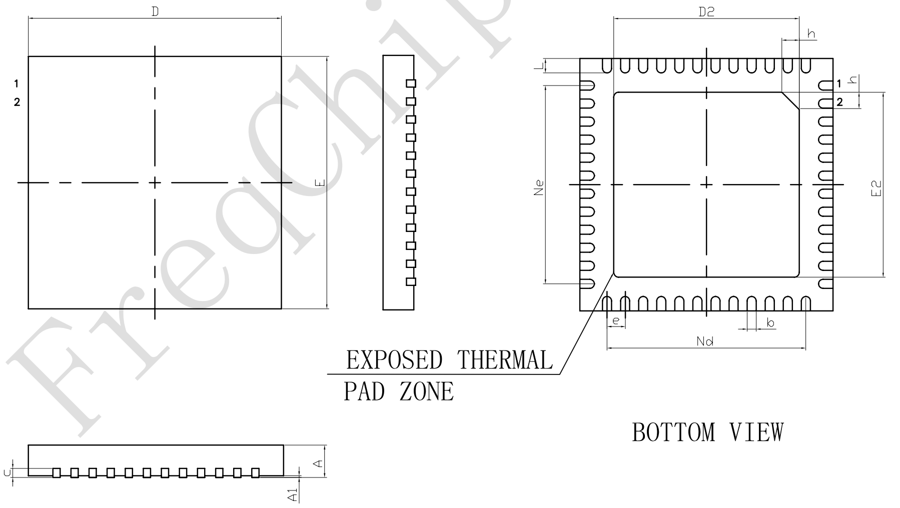
2.2.1 FR8003A/FR8003D



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0. 70	0.75	0. 80
A1	0	0. 02	0. 05
b	0. 15	0. 20	0. 25
b1	0. 14REF		
c	0. 203REF		
D	2. 90	3. 00	3. 10
D2	1. 80	1. 90	2. 00
e	0. 40BSC		
Ne	1. 60BSC		
Nd	1. 60BSC		
E	2. 90	3. 00	3. 10
E2	1. 80	1. 90	2. 00
L	0. 20	0. 25	0. 30
h	0. 25	0. 30	0. 35
K	0. 30REF		

Figure 2-7 FR8003A/8003D package physical dimensions

2.2.2 FR8008A

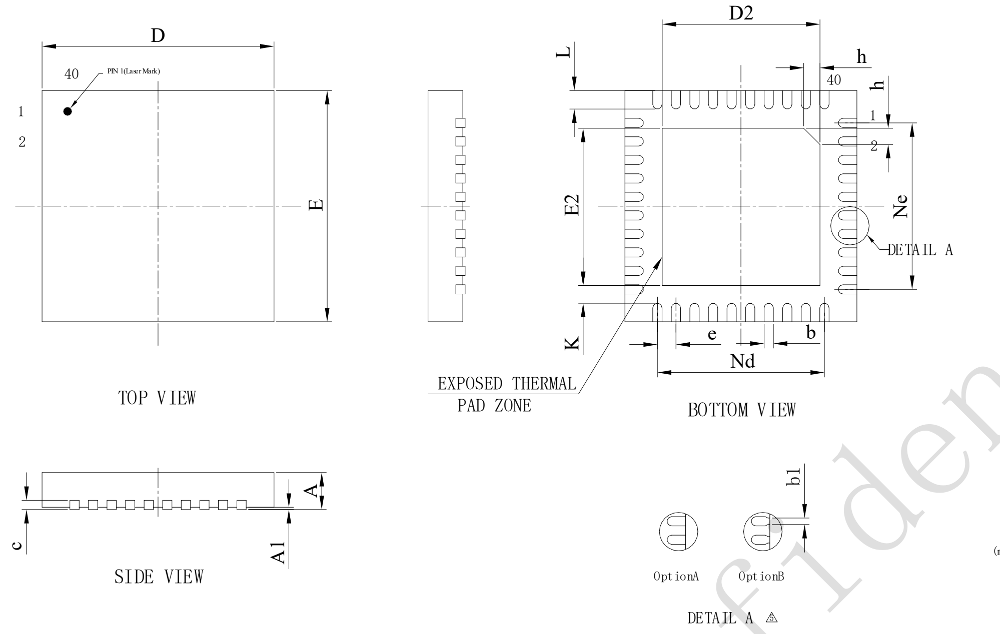


SYMBOL	MILLIMETER			
	MIN	NOM	MAX	
A	0. 70	0.75	0. 80	
A1	—	0.02	0. 05	
b	0. 15	0.20	0. 25	1
c	0. 18	0.20	0. 23	
D	5. 90	6. 00	6. 10	
D2	4. 10	4.20	4. 30	2
e	0. 40BSC			
Ne	4. 40BSC			
Nd	4. 40BSC			
E	5. 90	6. 00	6. 10	
E2	4. 10	4.20	4. 30	2
L	0. 35	0.40	0. 45	
h	0. 30	0. 35	0. 40	
L/F载体尺寸（MIL）	177*177			

Figure 2-8 FR8008A package physical dimensions

2.2.3 FR8008XP(FR8008AP/FR8008FP/FR8008GP-P/FR8008HP) /FR8008G

Note: The size of FR8008AP/FR8008G is 5.0 × 5.0 × 0.75, others are 5.0 × 5.0× 0.90, unit: mm



SYMBOL	MILLIMETER MIN	MILLIMETER NOM	MILLIMETER MAX	
A	0.70	0.75	0.80	
	0.80	0.85	0.90	3
	0.85	0.90	0.95	4
A1	—	0.02	0.05	
b	0.15	0.20	0.25	
b1	0.14REF			
c	0.18	0.20	0.25	
D	4.90	5.00	5.10	
D2	3.30	3.40	3.50	
e	0.40BSC			
Nd	3.60BSC			
E	4.90	5.00	5.10	
E2	3.30	3.40	3.50	
Ne	3.60BSC			
L	0.35	0.40	0.45	
K	0.20	—	—	
h	0.30	0.35	0.40	
L/F载体尺寸 (mil)	150*150			

Figure 2-9 FR8008XP package physical dimensions

2.3 Pins Description

FR800x series are CMOS devices. Floating level on input signals will cause unstable device operation and abnormal current consumption. Pull-up or Pull-down resistors should be used appropriately for input or bidirectional pins.

Table 2-1 pin abbreviations

Notation	Description
AI	Analog input
AO	Analog output
I/O	Bidirectional(digital)
PWR	Power
GND	Ground

2.3.1 FR8003A Pins

table 2-2 FR8003A pin description

Pin No.	Pin Name	Type	Description
1	BFB	AI	BUCK feedback input

Pin No.	Pin Name	Type	Description
2	HD_ONKEY	AI	Wake up input pin while in power off mode
3	RF	AI/O	RF input and output
4	XTALI	AI	24MHz Crystal oscillator input
5	XTALO	AO	24MHz Crystal oscillator output
6	PA5	I/O	PA5 / I2C0.DAT / SPIM0.IO3 / SPIS.CSN / UTXD0 / USBDM / PWM5 / PDM.DAT / ANT.TX / SIRIN / I2S.FRM
7	PA4	I/O	PA4 / I2C0.CLK / SPIM0.IO2 / SPIS.CLK / URXD0 / USBDP / PWM4 / PDM.CLK / CLK.OUT / I2S.BCLK
8	PA1	I/O	PA1 / I2C0.DAT / SPIM0.CSN / SPIS.CSN / UTXD0 / USBDM / PWM1 / PDM.DAT / SIROUT / I2S.FRM
9	PA0	I/O	PA0 / I2C0.CLK / SPIM0.CLK / SPIS.CLK / URXD0 / USBDP / PWM0 / PDM.CLK / SIRIN / I2S.BCLK
10	PC7	I/O	PC7 / I2C1.DAT / SPIM1.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / SWDIO / RS485.EN / I2S.MISO
11	PC6	I/O	PC6 / I2C1.CLK / SPIM1.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / SWTCK / SIROUT / I2S.MOSI
12	PC3	I/O	PC3 / I2C1.DAT / SPIM1.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / SWV / I2S.MISO
13	PC2	I/O	PC2 / I2C1.CLK / SPIM1.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / SWV / RS485.EN / I2S.MOSI
14	PD6	I/O	PD6 / I2C1.CLK / SPIM1.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / SARADC6 / SIROUT / I2S.MOSI
15	PD7	I/O	PD7 / I2C1.DAT / SPIM1.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / SARADC7 / RS485.EN / I2S.MISO
16	PD1	I/O	PD1 / I2C0.DAT / SPIM1.CSN / SPIS.CSN / UTXD0 / PWM1 / PDM.DAT / SARADC1 / SIROUT / I2S.FRM
17	PD0	I/O	PD0 / I2C0.CLK / SPIM1.CLK / SPIS.CLK / URXD0 / PWM0 / PDM.CLK / SARADC0 / SIRIN / I2S.BCLK
18	VDDIO	PWR	Power supply for IO
19	VCC	PWR	Power supply for chip
20	BSW	AO	BUCK Switch output

2.3.2 FR8003D Pins

table 2-3 FR8003D pin description

Pin No.	Pin Name	Type	Description
1	BFB	AI	BUCK feedback input
2	NC	-	Not connected
3	RF	AI/O	RF input and output
4	XTALI	AI	24MHz Crystal oscillator input
5	XTALO	AO	24MHz Crystal oscillator output
6	PA5	I/O	PA5 / I2C0.DAT / SPIM0.IO3 / SPIS.CSN / UTXD0 / USBDM / PWM5 / PDM.DAT / ANT.TX / SIRIN / I2S.FRM
7	PA4	I/O	PA4 / I2C0.CLK / SPIM0.IO2 / SPIS.CLK / URXD0 / USBDP / PWM4 / PDM.CLK / CLK.OUT / I2S.BCLK
8	PA1	I/O	PA1 / I2C0.DAT / SPIM0.CSN / SPIS.CSN / UTXD0 / USBDM / PWM1 / PDM.DAT / SIROUT / I2S.FRM
9	PA0	I/O	PA0 / I2C0.CLK / SPIM0.CLK / SPIS.CLK / URXD0 / USBDP / PWM0 / PDM.CLK / SIRIN / I2S.BCLK
10	PC7	I/O	PC7 / I2C1.DAT / SPIM1.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / SWDIO / RS485.EN / I2S.MISO
11	PC6	I/O	PC6 / I2C1.CLK / SPIM1.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / SWTCK / SIROUT / I2S.MOSI
12	PC3	I/O	PC3 / I2C1.DAT / SPIM1.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / SWV / I2S.MISO
13	PC2	I/O	PC2 / I2C1.CLK / SPIM1.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / SWV / RS485.EN / I2S.MOSI
14	PD6	I/O	PD6 / I2C1.CLK / SPIM1.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / SARADC6 / SIROUT / I2S.MOSI
15	PD7	I/O	PD7 / I2C1.DAT / SPIM1.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / SARADC7 / RS485.EN / I2S.MISO
16	PD1	I/O	PD1 / I2C0.DAT / SPIM1.CSN / SPIS.CSN / UTXD0 / PWM1 / PDM.DAT / SARADC1 / SIROUT / I2S.FRM
17	PD0	I/O	PD0 / I2C0.CLK / SPIM1.CLK / SPIS.CLK / URXD0 / PWM0 / PDM.CLK / SARADC0 / SIRIN / I2S.BCLK
18	VDDIO	PWR	Power supply for IO
19	VCC	PWR	Power supply for chip
20	BSW	AO	BUCK switch output

2.3.3 FR8008A Pins

table 2-4 FR8008A pin description

Pin No.	Pin Name	Type	Description
1	XO_32K	AO	32KHz Crystal oscillator output
2	XI_32K	AI	32KHz Crystal oscillator input
3	RSTN	AI	Chip reset pin
4	LDO_OUT	PWR	Linear regulator output with on/off switch control
5	VDDIO	PWR	Power supply for IO
6	VCC	PWR	Power supply for chip
7	BSW	AO	BUCK switch output
8	BFB	AI	BUCK feedback input
9	NC	-	Not connected
10	RF	AI/O	RF input and output
11	XO_24M	AO	24MHz Crystal oscillator output
12	XI_24M	AI	24MHz Crystal oscillator input
13	PA7	I/O	PA7 / I2C1.DAT / SPIM0.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / CLK.OUT / RS485.EN / I2S.MISO
14	PA6	I/O	PA6 / I2C1.CLK / SPIM0.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / ANT.RX / SIROUT / I2S.MOSI
15	PA5	I/O	PA5 / I2C0.DAT / SPIM0.IO3 / SPIS.CSN / UTXD0 / USBDM / PWM5 / PDM.DAT / ANT.TX / SIRIN / I2S.FRM
16	PA4	I/O	PA4 / I2C0.CLK / SPIM0.IO2 / SPIS.CLK / URXD0 / USBDP / PWM4 / PDM.CLK / CLK.OUT / I2S.BCLK
17	PA3	I/O	PA3 / I2C1.DAT / SPIM0.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / WLAN.RX / I2S.MISO
18	PA2	I/O	PA2 / I2C1.CLK / SPIM0.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / WLAN.TX / RS485.EN / I2S.MOSI
19	PA1	I/O	PA1 / I2C0.DAT / SPIM0.CSN / SPIS.CSN / UTXD0 / USBDM / PWM1 / PDM.DAT / SIROUT / I2S.FRM
20	PA0	I/O	PA0 / I2C0.CLK / SPIM0.CLK / SPIS.CLK / URXD0 / USBDP / PWM0 / PDM.CLK / SIRIN / I2S.BCLK
21	PB7	I/O	PB7 / I2C1.DAT / SPIM0.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / CLK.OUT / RS485.EN / I2S.MISO
22	PB6	I/O	PB6 / I2C1.CLK / SPIM0.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / ANT.RX / SIROUT / I2S.MOSI
23	PB5	I/O	PB5 / I2C0.DAT / SPIM0.IO3 / SPIS.CSN / UTXD0 / USBDM / PWM5 / PDM.DAT / ANT.TX / SIRIN / I2S.FRM

Pin No.	Pin Name	Type	Description
24	PB4	I/O	PB4 / I2C0.CLK / SPIM0.IO2 / SPIS.CLK / URXD0 / USBDP / PWM4 / PDM.CLK / CLK.OUT / LCD.DCX / I2S.BCLK
25	PB3	I/O	PB3 / I2C1.DAT / SPIM0.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / WLAN.RX / LCD.CSX / BURN.SPIMISO
26	PB2	I/O	PB2 / I2C1.CLK / SPIM0.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / WLAN.TX / RS485.EN / BURN.SPIMOSI
27	PB1	I/O	PB1 / I2C0.DAT / SPIM0.CSN / SPIS.CSN / UTXD0 / USBDM / PWM1 / PDM.DAT / SIROUT / BURN.SPICSN
28	PB0	I/O	PB0 / I2C0.CLK / SPIM0.CLK / SPIS.CLK / URXD0 / USBDP / PWM0 / PDM.CLK / SIRIN / BURN.SPICLK
29	PC7	I/O	PC7 / I2C1.DAT / SPIM1.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / SWDIO / RS485.EN / I2S.MISO
30	PC6	I/O	PC6 / I2C1.CLK / SPIM1.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / SWTCK / SIROUT / I2S.MOSI
31	PC5	I/O	PC5 / I2C0.DAT / SPIM1.IO3 / SPIS.CSN / UTXD0 / LCD.RDX / PWM5 / PDM.DAT / SWV / SIRIN / I2S.FRM / PSRAM.IO0 / LCD.D13
32	PC4	I/O	PC4 / I2C0.CLK / SPIM1.IO2 / SPIS.CLK / URXD0 / LCD.WRX / PWM4 / PDM.CLK / ANT.RX / LCD.TE / I2S.BCLK / PSRAM.IO2 / LCD.D12
33	PC3	I/O	PC3 / I2C1.DAT / SPIM1.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / SWV / I2S.MISO
34	PC2	I/O	PC2 / I2C1.CLK / SPIM1.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / SWV / RS485.EN / I2S.MOSI
35	PC1	I/O	PC1 / I2C0.DAT / SPIM1.CSN / SPIS.CSN / UTXD0 / LCD.DCX / PWM1 / PDM.DAT / SWV / SIROUT / I2S.FRM / PSRAM.CLK / LCD.D9
36	PC0	I/O	PC0 / I2C0.CLK / SPIM1.CLK / SPIS.CLK / URXD0 / LCD.CSX / PWM0 / PDM.CLK / SWV / SIRIN / I2S.BCLK / PSRAM.IO3 / LCD.D8
37	PE1	I/O	PE1 / I2C0.DAT / SPIM1.IO1 / SPIS.CSN / UTXD0 / UTXD1 / PWM1 / PDM.DAT / SIROUT / USBDM
38	PE0	I/O	PE0 / I2C0.CLK / SPIM1.IO0 / SPIS.CLK / URXD0 / URXD1 / PWM0 / PDM.CLK / SIRIN / USBDP
39	PD7	I/O	PD7 / I2C1.DAT / SPIM1.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / SARADC7 / RS485.EN / I2S.MISO
40	PD6	I/O	PD6 / I2C1.CLK / SPIM1.CLK / SPIS.MOSI / URTS0 / URXD1 /

Pin No.	Pin Name	Type	Description
			PWM6 / PDM.CLK / SARADC6 / SIROUT / I2S.MOSI
41	PD5	I/O	PD5 / I2C0.DAT / SPIM1.IO3 / SPIS.CSN / UTXD0 / PWM5 / PDM.DAT / SARADC5 / SIRIN / I2S.FRM
42	PD4	I/O	PD4 / I2C0.CLK / SPIM1.IO2 / SPIS.CLK / URXD0 / PWM4 / PDM.CLK / SARADC4 / I2S.BCLK
43	PD3	I/O	PD3 / I2C1.DAT / SPIM1.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / SARADC3 / I2S.MISO
44	PD2	I/O	PD2 / I2C1.CLK / SPIM1.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / SARADC2 / RS485.EN / I2S.MOSI
45	PD1	I/O	PD1 / I2C0.DAT / SPIM1.CSN / SPIS.CSN / UTXD0 / PWM1 / PDM.DAT / SARADC1 / SIROUT / I2S.FRM
46	PD0	I/O	PD0 / I2C0.CLK / SPIM1.CLK / SPIS.CLK / URXD0 / PWM0 / PDM.CLK / SARADC0 / SIRIN / I2S.BCLK
47	USB_DM	I/O	USB Digital Minus
48	USB_DP	I/O	USB Digital Positive

2.3.4 FR8008G

table 2-5 FR8008G pin description

Pin No.	Pin Name	Type	Description
1	DLDO_OUT	PWR	Linear regulator output with on/off switch control
2	RSTN	AI	Chip reset pin
3	VDDIO	PWR	Power supply for IO
4	VCC	PWR	Power supply for chip
5	BSW	AO	BUCK switch output
6	BFB	AI	BUCK feedback input
7	NC	-	Not connected
8	RF	AI/O	RF input and output
9	XO_24M	AO	24MHz Crystal oscillator output
10	XI_24M	AI	24MHz Crystal oscillator input
11	PA7	I/O	PA7 / I2C1.DAT / SPIM0.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / CLK.OUT / RS485.EN / I2S.MISO
12	PA6	I/O	PA6 / I2C1.CLK / SPIM0.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / ANT.RX / SIROUT / I2S.MOSI
13	PA5	I/O	PA5 / I2C0.DAT / SPIM0.IO3 / SPIS.CSN / UTXD0 / USBDM / PWM5 / PDM.DAT / ANT.TX / SIRIN / I2S.FRM
14	PA4	I/O	PA4 / I2C0.CLK / SPIM0.IO2 / SPIS.CLK / URXD0 / USBDP /

Pin No.	Pin Name	Type	Description
			PWM4 / PDM.CLK / CLK.OUT / I2S.BCLK
15	PA3	I/O	PA3 / I2C1.DAT / SPIM0.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / WLAN.RX / I2S.MISO
16	PA2	I/O	PA2 / I2C1.CLK / SPIM0.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / WLAN.TX / RS485.EN / I2S.MOSI
17	PA1	I/O	PA1 / I2C0.DAT / SPIM0.CSN / SPIS.CSN / UTXD0 / USBDM / PWM1 / PDM.DAT / SIROUT / I2S.FRM
18	PA0	I/O	PA0 / I2C0.CLK / SPIM0.CLK / SPIS.CLK / URXD0 / USBDP / PWM0 / PDM.CLK / SIRIN / I2S.BCLK
19	PB7	I/O	PB7 / I2C1.DAT / SPIM0.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / CLK.OUT / RS485.EN / I2S.MISO
20	PB6	I/O	PB6 / I2C1.CLK / SPIM0.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / ANT.RX / SIROUT / I2S.MOSI
21	PB5	I/O	PB5 / I2C0.DAT / SPIM0.IO3 / SPIS.CSN / UTXD0 / USBDM / PWM5 / PDM.DAT / ANT.TX / SIRIN / I2S.FRM
22	PB4	I/O	PB4 / I2C0.CLK / SPIM0.IO2 / SPIS.CLK / URXD0 / USBDP / PWM4 / PDM.CLK / CLK.OUT / LCD.DCX / I2S.BCLK
23	PB3	I/O	PB3 / I2C1.DAT / SPIM0.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / WLAN.RX / LCD.CSX / BURN.SPIMISO
24	PB2	I/O	PB2 / I2C1.CLK / SPIM0.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / WLAN.TX / RS485.EN / BURN.SPIMOSI
25	PB1	I/O	PB1 / I2C0.DAT / SPIM0.CSN / SPIS.CSN / UTXD0 / USBDM / PWM1 / PDM.DAT / SIROUT / BURN.SPICSN
26	PB0	I/O	PB0 / I2C0.CLK / SPIM0.CLK / SPIS.CLK / URXD0 / USBDP / PWM0 / PDM.CLK / SIRIN / BURN.SPICLK
27	PC7	I/O	PC7 / I2C1.DAT / SPIM1.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / SWDIO / RS485.EN / I2S.MISO
28	PC6	I/O	PC6 / I2C1.CLK / SPIM1.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / SWTCK / SIROUT / I2S.MOSI
29	PE1	I/O	PE1 / I2C0.DAT / SPIM1.IO1 / SPIS.CSN / UTXD0 / UTXD1 / PWM1 / PDM.DAT / SIROUT / USBDM
30	PE0	I/O	PE0 / I2C0.CLK / SPIM1.IO0 / SPIS.CLK / URXD0 / URXD1 / PWM0 / PDM.CLK / SIRIN / USBDP
31	PD7	I/O	PD7 / I2C1.DAT / SPIM1.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / SARADC7 / RS485.EN / I2S.MISO
32	PD6	I/O	PD6 / I2C1.CLK / SPIM1.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / SARADC6 / SIROUT / I2S.MOSI

Pin No.	Pin Name	Type	Description
33	PD5	I/O	PD5 / I2C0.DAT / SPIM1.IO3 / SPIS.CSN / UTXD0 / PWM5 / PDM.DAT / SARADC5 / SIRIN / I2S.FRM
34	PD4	I/O	PD4 / I2C0.CLK / SPIM1.IO2 / SPIS.CLK / URXD0 / PWM4 / PDM.CLK / SARADC4 / I2S.BCLK
35	PD3	I/O	PD3 / I2C1.DAT / SPIM1.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / SARADC3 / I2S.MISO
36	PD2	I/O	PD2 / I2C1.CLK / SPIM1.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / SARADC2 / RS485.EN / I2S.MOSI
37	PD1	I/O	PD1 / I2C0.DAT / SPIM1.CSN / SPIS.CSN / UTXD0 / PWM1 / PDM.DAT / SARADC1 / SIROUT / I2S.FRM
38	PD0	I/O	PD0 / I2C0.CLK / SPIM1.CLK / SPIS.CLK / URXD0 / PWM0 / PDM.CLK / SARADC0 / SIRIN / I2S.BCLK
39	XO_32K	AO	32KHz Crystal oscillator output
40	XI_32K	AI	32KHz Crystal oscillator input

2.3.5 FR8008XP

table 2-6 FR8008XP pin description

Pin No.	Pin Name	Type	Description
1	DLDO_OUT	PWR	Linear regulator output with on/off switch control
2	RSTN	AI	Chip reset pin
3	VDDIO	PWR	Power supply for IO
4	VCC	PWR	Power supply for chip
5	BSW	AO	BUCK switch output
6	BFB	AI	BUCK feedback input
7	NC	-	Not connected
8	RF	AI/O	RF input and output
9	XO_24M	AO	24MHz Crystal oscillator output
10	XI_24M	AI	24MHz Crystal oscillator input
11	PA7	I/O	PA7 / I2C1.DAT / SPIM0.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / CLK.OUT / RS485.EN / I2S.MISO
12	PA6	I/O	PA6 / I2C1.CLK / SPIM0.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / ANT.RX / SIROUT / I2S.MOSI
13	PA5	I/O	PA5 / I2C0.DAT / SPIM0.IO3 / SPIS.CSN / UTXD0 / USBDM / PWM5 / PDM.DAT / ANT.TX / SIRIN / I2S.FRM
14	PA4	I/O	PA4 / I2C0.CLK / SPIM0.IO2 / SPIS.CLK / URXD0 / USBDP /

Pin No.	Pin Name	Type	Description
			PWM4 / PDM.CLK / CLK.OUT / I2S.BCLK
15	PA3	I/O	PA3 / I2C1.DAT / SPIM0.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / WLAN.RX / I2S.MISO
16	PA2	I/O	PA2 / I2C1.CLK / SPIM0.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / WLAN.TX / RS485.EN / I2S.MOSI
17	PA1	I/O	PA1 / I2C0.DAT / SPIM0.CSN / SPIS.CSN / UTXD0 / USBDM / PWM1 / PDM.DAT / SIROUT / I2S.FRM
18	PA0	I/O	PA0 / I2C0.CLK / SPIM0.CLK / SPIS.CLK / URXD0 / USBDP / PWM0 / PDM.CLK / SIRIN / I2S.BCLK
19	PB7	I/O	PB7 / I2C1.DAT / SPIM0.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / CLK.OUT / RS485.EN / I2S.MISO
20	PB6	I/O	PB6 / I2C1.CLK / SPIM0.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / ANT.RX / SIROUT / I2S.MOSI
21	PB5	I/O	PB5 / I2C0.DAT / SPIM0.IO3 / SPIS.CSN / UTXD0 / USBDM / PWM5 / PDM.DAT / ANT.TX / SIRIN / I2S.FRM
22	PB4	I/O	PB4 / I2C0.CLK / SPIM0.IO2 / SPIS.CLK / URXD0 / USBDP / PWM4 / PDM.CLK / CLK.OUT / LCD.DCX / I2S.BCLK
23	PB3	I/O	PB3 / I2C1.DAT / SPIM0.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / WLAN.RX / LCD.CSX / BURN.SPIMISO
24	PB2	I/O	PB2 / I2C1.CLK / SPIM0.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / WLAN.TX / RS485.EN / BURN.SPIMOSI
25	PB1	I/O	PB1 / I2C0.DAT / SPIM0.CSN / SPIS.CSN / UTXD0 / USBDM / PWM1 / PDM.DAT / SIROUT / BURN.SPICSN
26	PB0	I/O	PB0 / I2C0.CLK / SPIM0.CLK / SPIS.CLK / URXD0 / USBDP / PWM0 / PDM.CLK / SIRIN / BURN.SPICLK
27	PC7	I/O	PC7 / I2C1.DAT / SPIM1.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / SWDIO / RS485.EN / I2S.MISO
28	PC6	I/O	PC6 / I2C1.CLK / SPIM1.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / SWTCK / SIROUT / I2S.MOSI
29	PE1	I/O	PE1 / I2C0.DAT / SPIM1.IO1 / SPIS.CSN / UTXD0 / UTXD1 / PWM1 / PDM.DAT / SIROUT / USBDM
30	PE0	I/O	PE0 / I2C0.CLK / SPIM1.IO0 / SPIS.CLK / URXD0 / URXD1 / PWM0 / PDM.CLK / SIRIN / USBDP
31	PD7	I/O	PD7 / I2C1.DAT / SPIM1.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / SARADC7 / RS485.EN / I2S.MISO
32	PD6	I/O	PD6 / I2C1.CLK / SPIM1.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / SARADC6 / SIROUT / I2S.MOSI

Pin No.	Pin Name	Type	Description
33	PD5	I/O	PD5 / I2C0.DAT / SPIM1.IO3 / SPIS.CSN / UTXD0 / PWM5 / PDM.DAT / SARADC5 / SIRIN / I2S.FRM
34	PD4	I/O	PD4 / I2C0.CLK / SPIM1.IO2 / SPIS.CLK / URXD0 / PWM4 / PDM.CLK / SARADC4 / I2S.BCLK
35	PD3	I/O	PD3 / I2C1.DAT / SPIM1.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / SARADC3 / I2S.MISO
36	PD2	I/O	PD2 / I2C1.CLK / SPIM1.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / SARADC2 / RS485.EN / I2S.MOSI
37	PD1	I/O	PD1 / I2C0.DAT / SPIM1.CSN / SPIS.CSN / UTXD0 / PWM1 / PDM.DAT / SARADC1 / SIROUT / I2S.FRM
38	PD0	I/O	PD0 / I2C0.CLK / SPIM1.CLK / SPIS.CLK / URXD0 / PWM0 / PDM.CLK / SARADC0 / SIRIN / I2S.BCLK
39	XTALO_32K (FR8008XP)	AO	32KHz Crystal oscillator output
	USB_DM (FR8008AP)	I/O	USB Digital Minus
40	XTALI_32K (FR8008XP)	AI	32KHz Crystal oscillator input
	USB_DP (FR8008AP)	I/O	USB Digital Positive

2.4 Application Circuit

2.4.1 FR8003A Circuit

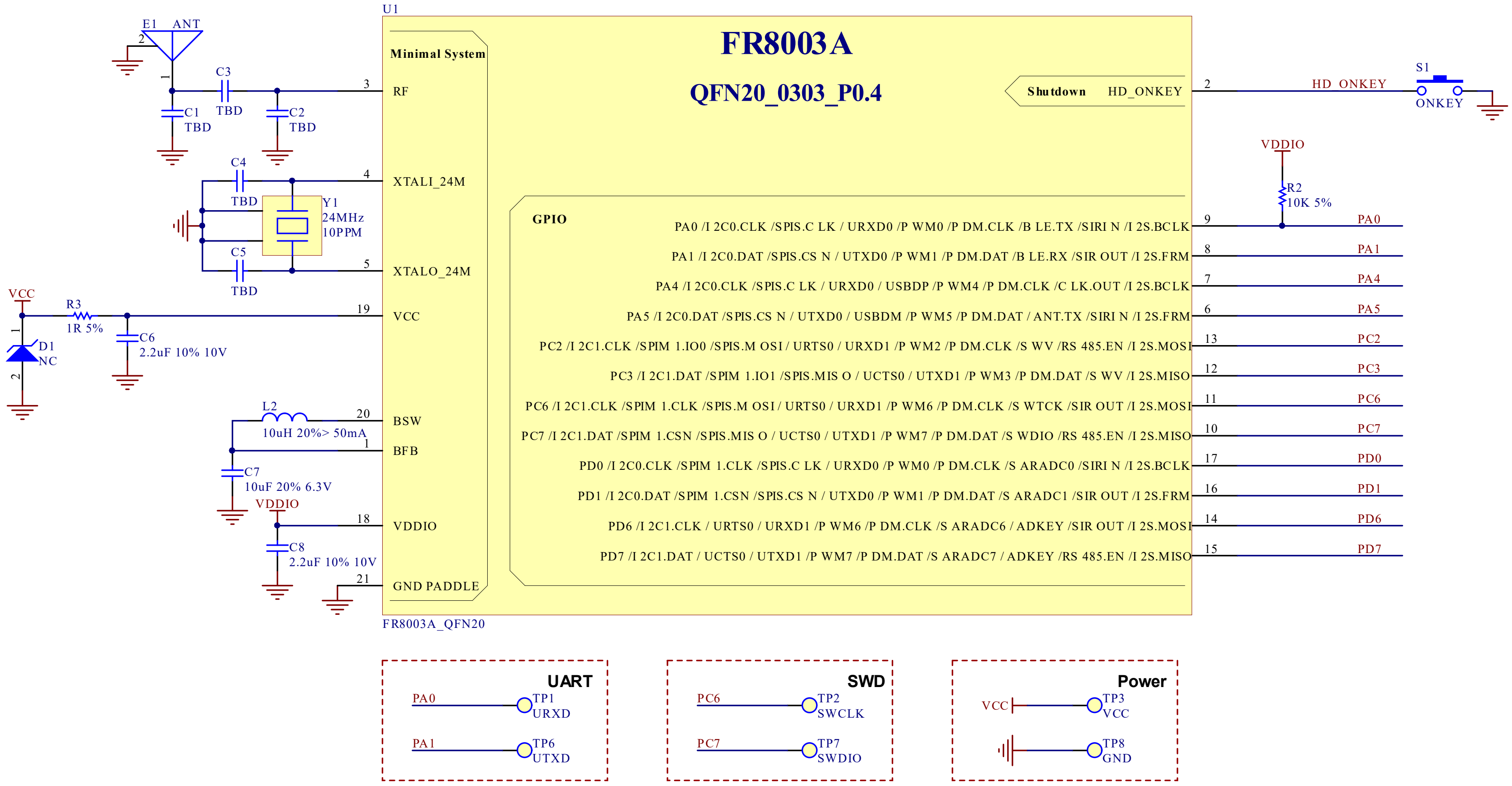


Figure 2-10 FR8003A application circuit

2.4.2 FR8003D Circuit

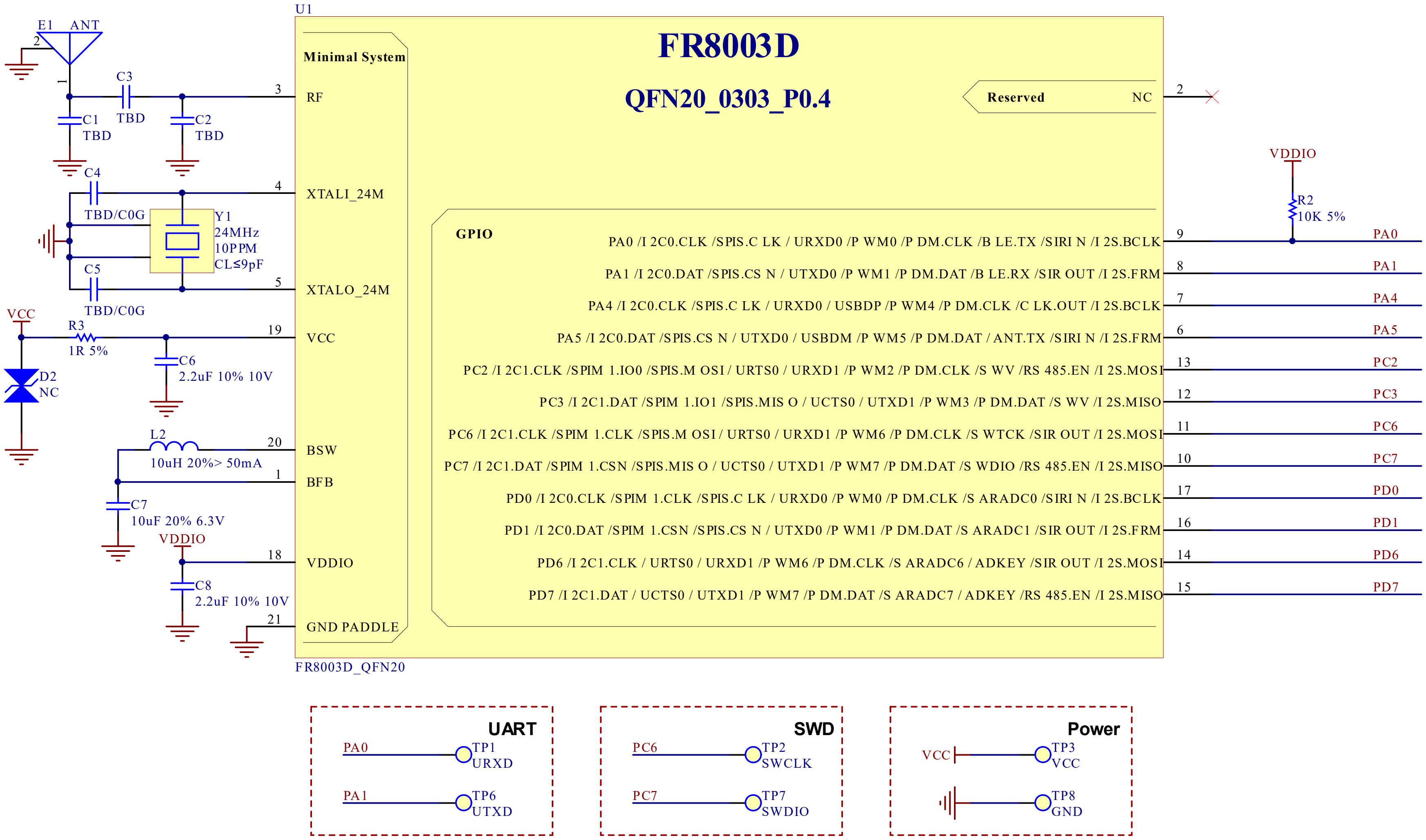


Figure 2-11 FR8003D application circuit

2.4.3 FR8008A Circuit

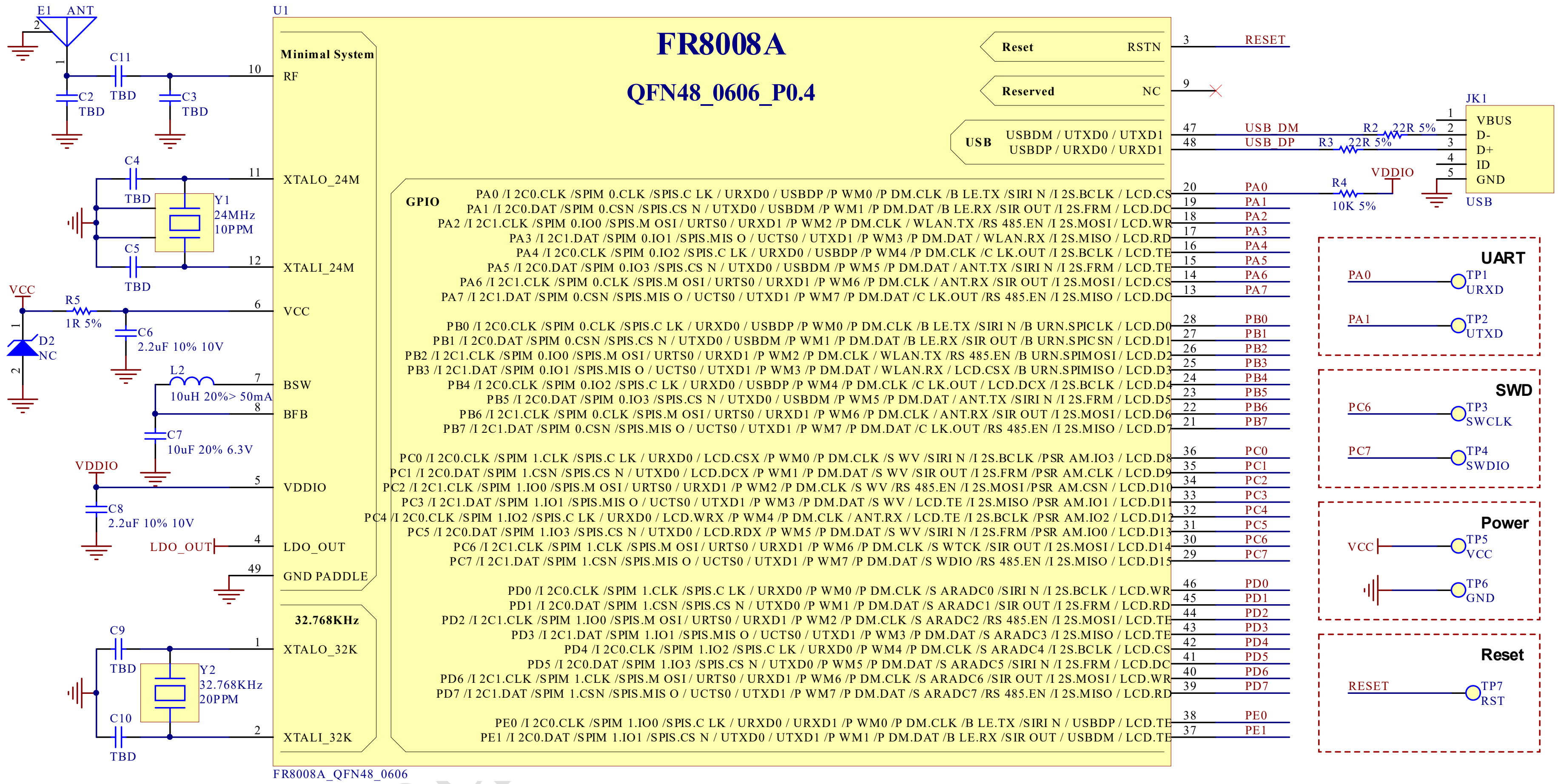


Figure 2-12 FR8008A application circuit

2.4.4 FR8008G Circuit

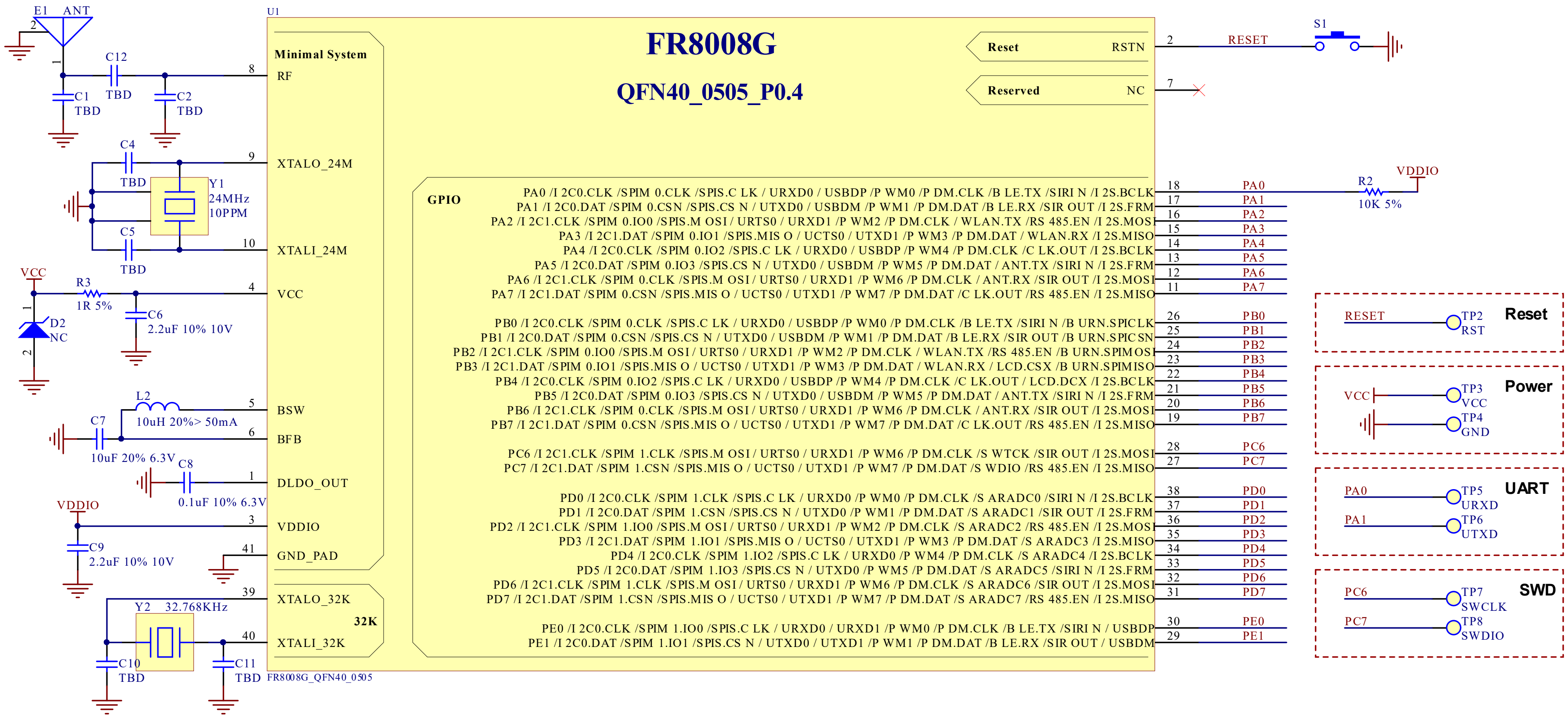


Figure 2-13 FR8008G application circuit

2.4.5 FR8008XP Circuit

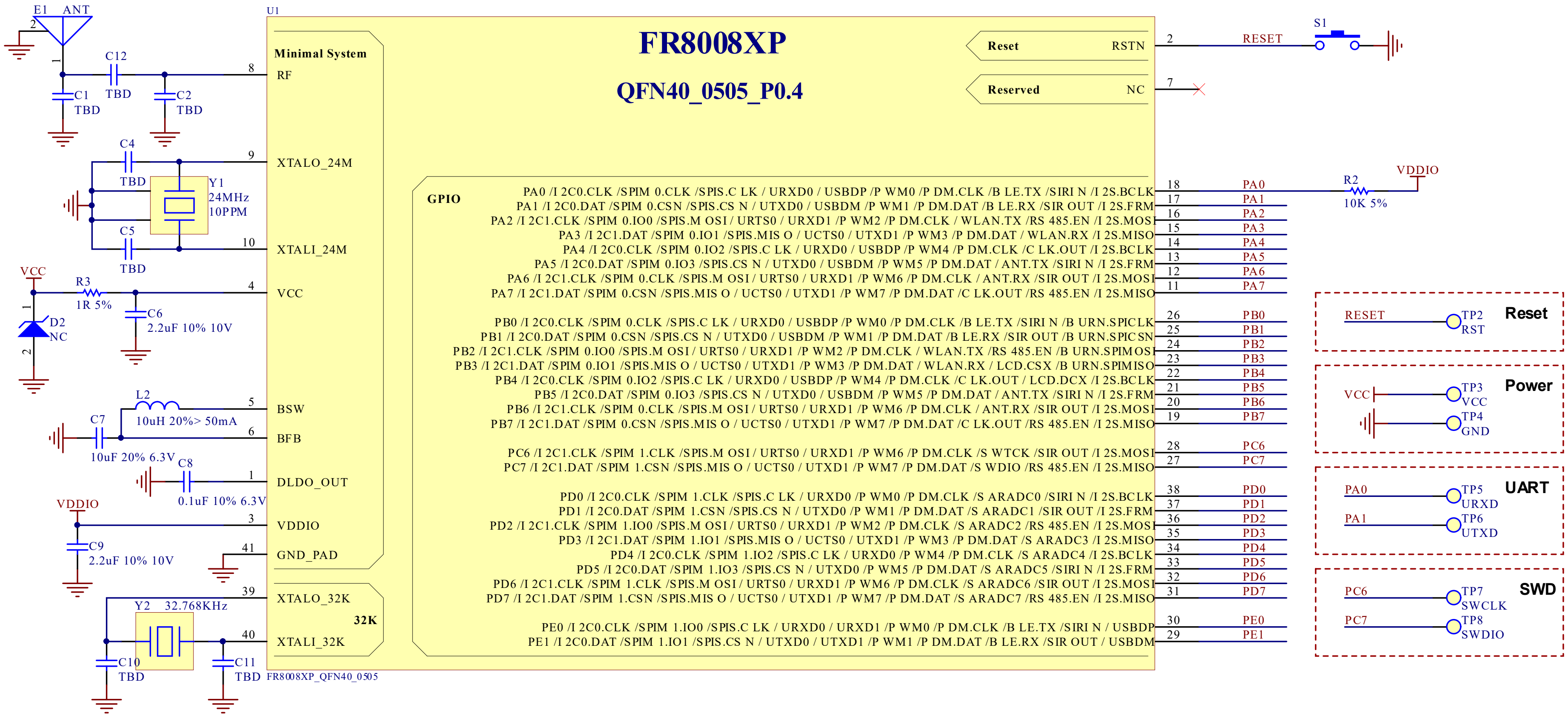


Figure 2-14 FR8008XP application circuit

2.4.6 FR8008AP Circuit

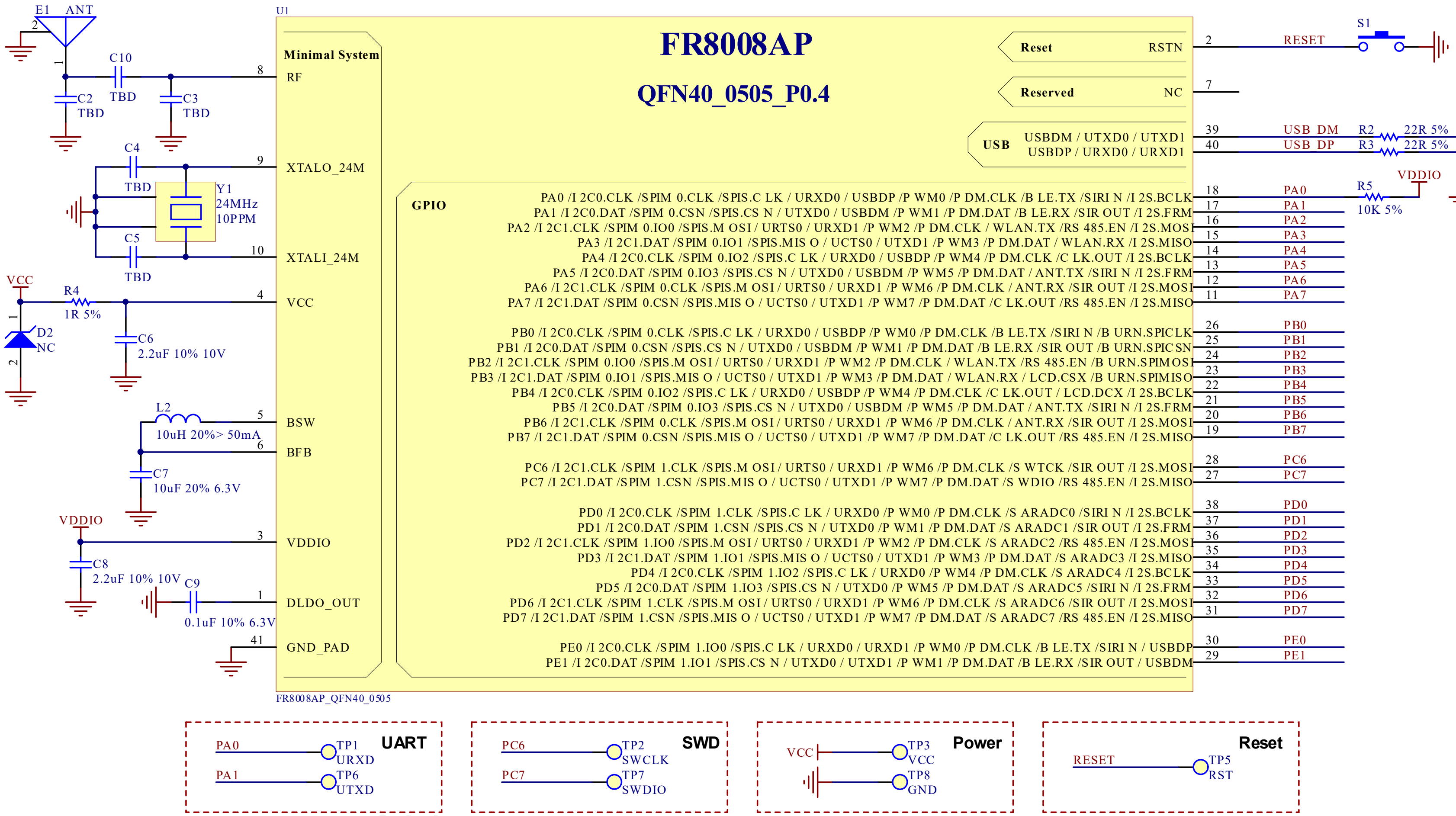


Figure 2-15 FR8008AP application circuit

3. Electrical Characteristics

3.1 Absolute Maximum Ratings

Continuous operation at or beyond these conditions may permanently damage the device.

Table 3-1 Absolute Maximum Ratings

Rating		**Min**	**Max**	**Unit**
Storage Temperature		-40	125	℃
Core Supply Voltage		0.9	1.2	V
I/O Voltage	VDDIO	1.8	3.6	V
Supply Voltage	VCC	2.0 2.7（FR8008XP）	3.6	V

3.2 Recommended Operating Conditions

Table 3-2 Recommended Operating Conditions

Operating Condition		**Min**	**Typ**	**Max**	**Unit**
Operating Temperature Range		-40	25	85	℃
Core Supply Voltage		0.9	1.1	1.2	V
I/O Voltage	VDDIO	1.8	2.9	3.6	V
Supply Voltage	VCC	2.0 2.7（FR8008XP）	3.3	3.6	V

Note: VDDIO : default value(Power-on) is 1.8V, user configurable later

3.3 Power Consumption

Table 3-3 Power Consumption

Operation Mode	**Average**	**Max**	**Unit**
TX peek current (0dB)		6.2	mA
RX peek current		6.5	mA
Sleep mode current	＜12(include 56K retention RAM)	18	μA
	＜10(include 48K retention RAM)	15	μA
	＜9 (include 32K retention RAM)	13	μA
Power off mode current	＜3.5		μA
Shutdown mode current	＜80		nA

3.4 Crystal Oscillator

Table 3-4 Crystal Oscillator

Parameter	Min	Typ	Max	Unit
Clock Frequency	24	24	24	MHz
CL Load capacitance	-	9	9	pF
Tolerance	-	+/-10	-	PPM
Motional resistance	-	-	60	Ω
Shunt capacitance	-	-	2	pF

3.5 IO Resistor

Table 3-5 IO Resistor

Peripheral	Mode	VDD	Value
PA0~PA7 PB0~PB7 PC0~PC5 PD2~PD7	Pull-up	3.3V	10K
	Pull-down		5K
PC6~PC7 PD0~PD1	Pull-up	3.3V	5K
	Pull-down		
USB_DP USB_DM	Pull-up	3.3V	1.5K
	Pull-down		15K

3.6 ESD

Table 3-6 ESD

Pin name	Human body model(HBM)	Charged-device model(CDM)
RF	±2000V	±200V
XTALI	±2000V	±500V
XTALO	±2000V	±500V
OTHERS	±2000V	±500V

Abbreviations

Abbreviations	Descriptions
AEC	acoustic echo cancellers
AGC	Automatic Gain Control
ANS	Automatic Noise Suppression
ADC	Analog-to-Digital-Converter
DAC	Digital-to-Analog-Converter
GPIO	General Purpose Input Output
MIC	Microphone
PMU	Power Management Unit
OSC	Oscillator
PA	Power Amplifier
SoC	system on chip

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Revision History

Feedback:

Freqchip welcomes feedback on this product and this document. If you have comments or suggestions, please send an email to docs@freqchip.com.

Reversion Number	Reversion Date	Description
V0.2	2022.10.17	Initial Draft
V0.3	2022.12.29	Modify Cortex-M3 Feature
V0.3.1	2023.02.13	Modify VDDIO and VBAT value in Chapter 3
V0.3.2	2023.02.24	Add FR8008G
V0.3.5	2023.05.10	Delete VCHG
V0.3.15	2024.09.18	Modify Ordering Information
V0.3.16	2024.11.06	Add MSL Information
V0.3.17	2024.11.20	Add Crystal Oscillator and IO Resistor Information, and modify FR800XP VDD Add sleep mode current parameter
V0.3.18	2025.01.07	Modify Chapter 3.6 ESD Information
V0.3.19	2025.06.05	Modify Chapter 1.2 QSPI